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Lasso Penalization for High-Dimensional Beta Regression Models: Computation, Analysis, and Inference

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ABSTRACT

Beta regression is commonly employed when the outcome variable is a proportion. Since its conception, the approach has been widely used in applications spanning various scientific fields. A series of extensions have been proposed over time, several of which address variable selection and penalized estimation, e.g., with an ℓ_1-penalty (LASSO). However, a theoretical analysis of this popular approach in the context of Beta regression with high-dimensional predictors is lacking. In this paper, we aim to close this gap. A particular challenge arises from the non-convexity of the associated negative log-likelihood, which we address by resorting to a framework for analyzing stationary points in a neighborhood of the target parameter. Leveraging this framework, we derive a non-asymptotic bound on the ℓ_1-error of such stationary points. In addition, we propose a debiasing approach to construct confidence intervals for the regression parameters. A proximal gradient algorithm is devised for optimizing the resulting penalized negative log-likelihood function. Our theoretical analysis is corroborated via simulation studies, and a real data example concerning the prediction of county-level proportions of incarceration is presented to showcase the practical utility of our methodology.

1 | Introduction

Beta regression (Ferrari and Cribari-Neto 2004; Simas et al. 2010; Liu and Eugenio 2018) is a well-studied approach to regression modeling when the response variable is a proportion (e.g., county-level percentages of persons having health insurance, fraction of income spent on housing, proportion of land covered by forests), or more generally, contained in a bounded interval. Unlike proportions derived from binomial data, such responses are distinctly different from binomial response, which is typically modeled with a generalized linear model (McCullagh and Nelder 1989), in that numerator and/or denominator are either unknown, non-integer, or disregarded since a Binomial model may not appear appropriate for various reasons. For example, Binomial regression assumes that variances are inversely proportional to the number of trials so that populous counties would be treated differently from counties with a small population.

In this paper, we study Beta regression in a high-dimensional setup in which the number of predictors p can exceed the sample size n, potentially dramatically so. Generally speaking, it is well-known that consistent estimation in such a setup can be accomplished if the underlying regression parameter is sparse, i.e., if most of the entries of the target parameter are (approximately) zero (e.g., Bühlmann and van de Geer 2011). The use of sparsity-promoting regularization via a suitable penalty such as the ℓ_1-norm (aka LASSO (Tibshirani 1996)) has proven to be pivotal in this regard (e.g., Candès and Tao 2007; Bickel et al. 2009; Meinshausen and Yu 2009; Huang and Zhang 2012; Greenshtein and Ritov 2004; van de Geer 2008). One of the primary contributions of this paper is a non-asymptotic bound on the estimation error in ℓ_1-norm that is in line with that literature. The main challenge towards such a bound is the non-convexity of the objective function, which places the problem outside established frameworks (Negahban et al. 2012; Koltchinskii 2011; Wainwright 2019) relying on convexity.

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Instead, we adopt the framework developed in (Elsener and van de Geer 2018) geared towards showing consistency for stationary points of non-convex ℓ_1-penalized objectives in a neighborhood of the true parameter. The fact that the gradient of that objective depends on the log-response, which is unbounded, requires additional care in the analysis. We note that a variety of non-convex M-estimators have been studied in the literature (e.g., Loh and Wainwright 2012, 2015; Song et al. 2020; Mei et al. 2018). However, the body of work is scarce by comparison. The present paper adds to this literature.

In addition to the aforementioned ℓ_1-bound, we also provide methodology for constructing confidence intervals for low-dimensional parameters of the high-dimensional target, building on the debiasing approach pioneered in (Zhang and Zhang 2014) and further developed in (Javanmard and Montanari 2014; van de Geer et al. 2014). We present a scalable proximal gradient descent algorithm for minimizing the ℓ_1-penalized negative log-likelihood. We corroborate our theoretical analysis via simulation studies and conclude with an illustrative case study from criminal justice research.

Related work. Model selection for Beta regression is studied in (Bayer and Cribari-Neto 2015, 2017). Variable selection for additive models via boosting is proposed in (Schmid et al. 2013). Random forest is another widely used approach for fitting non-linear models while performing variable selection; a variant for Beta regression is developed in (Weinhold et al. 2020). Several papers discuss penalized estimation for Beta regression, via the ridge penalty or an ℓ_1-penalty as considered herein (Abonazel and Taha 2023; Firinguetti et al. 2024; Zhao et al. 2016; Zhao et al. 2014). The paper (Abonazel and Taha 2023) is dedicated to the use of a ridge penalty. The work (Zhao et al. 2014) considers a double penalty for both the means and the dispersion parameters, adopting the framework in Fan and Li (Fan and Li 2001), which includes non-convex penalties such as the smoothly clipped absolute deviation (SCAD). Accordingly, (Zhao et al. 2014) present an algorithm based on coordinate descent operating on a local quadratic approximation of the penalized log-likelihood, and demonstrate the oracle property of their estimators in the spirit of (Fan and Li 2001). In particular, the analysis in (Zhao et al. 2014) is limited to the low-dimensional case with a fixed number of predictors. A similar algorithm as in (Zhao et al. 2014) is presented in (Firinguetti et al. 2024) and evaluated via simulations. The paper (Zhao et al. 2016) proposes a quasi-likelihood approach to Beta regression in which the objective function has the same form as that of logistic regression and the mean function is expanded in a partially linear single index model. Variable selection in this setup based on penalties as in Fan and Li (Fan and Li 2001) is studied as well.

To our knowledge, there are no prior works containing an analysis of ℓ_1-penalized Beta regression in a high-dimensional setup. While finalizing the work on this manuscript, we became aware of the recent work (Stein et al. 2025) on sparse Beta regression models for network analysis. However, this work borrows its title from the Beta model in network analysis rather than the Beta distribution, and employs a logistic regression-style likelihood. Hence this work is not directly related to the present paper.

TABLE 1 | Summary of notation used repeatedly in this paper.

X	predictor variables	$\lambda = \lambda_n$	regularization parameter
Y	response variable	R_n	empirical risk
β^*	"true" regression vector	R	risk
ϕ^*	"true" scale parameter	$\mu(z)$	$\exp(z)/(1+\exp(z))$
β, ϕ	generic parameter values	$\mu(\beta)$	$\mu(x^\top\beta)$
∇, ∇^2	gradient, Hessian	$\mu'(\beta)$	$\frac{d\mu(z)}{dz}\Big\vert_{z=x^\top\beta}$
Γ	Gamma function	$\mu_i(\beta)$etc.	$\mu(X_i^\top\beta)$
Ψ	Digamma function	Beta(a,b)	Beta distribution

Outline. Our notation is summarized in the paragraph below. Our approach and its implementation are provided in §2. The main results of our theoretical analysis are presented in §3 and proved in the appendix. Empirical results in the form of simulations and the aforementioned case study are the subjects of §4. We conclude in §5.

Notation. Frequently used symbols are summarized in Table 1. In addition, for functions $f(n)$ and $g(n)$, we write $f(n) \lesssim g(n)$ to mean that $f(n) \le Cg(n)$ for a universal constant $C > 0$. Equivalently, $f(n) = O(g(n))$, where $O(\cdot)$ is the Landau big-O. If $f(n) \lesssim g(n)$ and $g(n) \lesssim f(n)$ hold simultaneously, we write $f(n) \asymp g(n)$. We write $C_{\dots}, C'_{\dots}, C''_{\dots}, \overline{C}_{\dots}, \tilde{C}_{\dots}, c_{\dots}$ etc. to denote positive constants depending at most on the list of variables $\dots$ in their subscripts. Their values may change from instance to instance.

2 | Methodology

2.1 | Approach

The approach to Beta regression adopted in this paper is in line with (Ferrari and Cribari-Neto 2004) and a corresponding implementation in R (Cribari-Neto and Zeileis 2010). In particular, we adopt the reparameterization in terms of a mean parameter μ and a scale parameter ϕ as opposed to the parameterization of the Beta distribution that is conventionally employed outside regression contexts. For the sake of reference, the probability density function of the $\mathsf{Beta}(a,b)$-distribution with parameters $a,\ b \ge 1$, is given by

$$f(x) = \frac{1}{\mathsf{B}(a,b)} x^{a-1} \cdot (1-x)^{b-1},\ x \in (0,1),$$

where $\mathsf{B}(\cdot)$ denotes the Beta function. Mean and variance are given by $\mu = \frac{a}{a+b}$ and $v = \frac{ab}{(a+b)^2(a+b+1)}$, respectively. Defining $\phi = a + b$, the variance can be re-expressed as

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$v = \mu(1-\mu)(\phi+1)^{-1}$. Accordingly, in a regression setup we postulate that

$$Y_i|X_i \sim \text{Beta}(\mu_i, \phi_i),\ \mu_i = \frac{\exp(\beta_0^* + X_i^\top \beta^*)}{1+\exp(\beta_0^* + X_i^\top \beta^*)},\ \phi_i \equiv \phi^*,\ \ 1 \le i \le n, \tag{1}$$

where the assumption of a homogeneous scale parameter $\phi_i^* \equiv \phi^*$, $1 \le i \le n$, is made for simplicity throughout this paper while acknowledging that having the scale parameter depend on covariates as well is a common option.

Letting $f_i(y)$, $y \in (0,1)$, denote the PDF of $Y_i|X_i$, we have

$$f_i(y) = \frac{\Gamma(\phi^*)}{\Gamma(\mu_i \cdot \phi^*)\Gamma((1-\mu_i)\phi^*)} y^{\mu_i\phi^*-1}(1-y)^{(1-\mu_i)\phi^*-1},\ 1 \le i \le n, \tag{2}$$

where we have used the well-known relationship $\mathsf{B}(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, $a, b > 0$.

Penalized Likelihood. According to (2), the negative log-likelihood for the parameters $\theta = (\beta_0, \beta, \phi)$ given $\{(X_i, Y_i)\}_{i=1}^n$ is obtained as

$$\begin{aligned} R_n(\theta) = -\log\Gamma(\phi) + \frac{1}{n}\sum_{i=1}^n \big\{ &\log\Gamma(\mu_i\phi) + \log\Gamma((1-\mu_i)) \\ &- (\mu_i\phi)\log Y_i - ((1-\mu_i)\phi - 1)\log(1-Y_i)\big\}, \end{aligned} \tag{3}$$

where $\mu_i = \mu_i(\beta_0, \beta) = \exp(\beta_0 + X_i^\top\beta)/(1+\exp(\beta_0 + X_i^\top\beta))$, $1 \le i \le n$. To estimate the parameters in the presence of high-dimensional predictors, we propose to minimize the ℓ_1-penalized counterpart to (3), i.e.,

$$\widehat{\theta} \in \arg\min_\theta \{R_n(\theta) + \lambda_n \|\beta\|_1\}, \tag{4}$$

where we note that the ℓ_1-penalty (aka lasso penalty) excludes β_0 and ϕ. Here, $\lambda_n > 0$ denotes the tuning parameter controlling the strength of the penalty.

De-biasing and confidence intervals. To construct confidence intervals for the regression parameters, we adopt the approach pioneered in (Zhang and Zhang 2014) and further developed in (Javanmard and Montanari 2014) and (van de Geer et al. 2014). The rationale behind this approach is as follows. From the Karush-Kuhn-Tucker (KKT) optimality conditions associated with the minimization problem in (4) it follows that

$$\nabla_{\beta_0,\beta} R_n(\widehat{\theta}) + \lambda_n \widehat{z} = 0,$$

where $\nabla_{\beta_0,\beta}$ denotes the gradient operator w.r.t. β_0 and β and $\widehat{z}$ is a vector whose entries are contained in $[-1,1]$. Using the Taylor approximation

$$\nabla_{\beta_0,\beta} R_n(\theta^*) \approx \nabla_{\beta_0,\beta} R_n(\widehat{\theta}) + \nabla^2_{\beta_0,\beta} R_n(\theta^*) \begin{pmatrix} \widehat{\beta}_0 - \beta_0^* \\ \widehat{\beta} - \beta^* \end{pmatrix}$$

and assuming that that the right hand side is close to zero (which is reasonable in light of the law of large numbers and the fact θ^* is the population minimizer), the de-biased estimator $(\widehat{\beta}_0^{\text{db}}, \widehat{\beta}^{\text{db}})$ is obtained as

$$\begin{pmatrix} \widehat{\beta}_0^{\text{db}} \\ \widehat{\beta}^{\text{db}} \end{pmatrix} = \begin{pmatrix} \widehat{\beta}_0 \\ \widehat{\beta} \end{pmatrix} + \lambda_n \widehat{\Omega}\widehat{z},$$

where $\widehat{\Omega}$ is an "approximate inverse" of $\nabla^2_{\beta_0,\beta} R_n(\theta^*) \approx \nabla^2_{\beta_0,\beta} R_n(\widehat{\beta})$ satisfying

$$\|\nabla^2_{\beta_0,\beta} R_n(\widehat{\theta})\widehat{\Omega} - I_{d+1}\|_\infty \lesssim \lambda_0, \qquad \lambda_0 = \|\nabla^2_{\beta_0,\beta} R_n(\widehat{\theta})\,\{\nabla^2_{\beta_0,\beta} R_n(\theta^*)\}^{-1}\|_\infty, \tag{5}$$

where λ_0 can be shown to be $O(\sqrt{\log(p)/n})$ in benign setups (Javanmard and Montanari 2014), and $\|\cdot\|_\infty$ here denotes the entry-wise ℓ_∞-norm. Following (van de Geer et al. 2014), we expect that under suitable conditions $(\widehat{\beta}_0^{\text{db}}, \widehat{\beta}^{\text{db}})$ are asymptotically normal, centered at (β_0^*, β^*) and covariance matrix estimable by $\frac{1}{n}\widehat{\Omega}\widehat{M}\widehat{\Omega}^\top$ with $\widehat{M} = \sum_{i=1}^n \nabla_{\beta_0,\beta} R_n^i(\widehat{\theta})^{\otimes^2}$, where R_n^i constitutes the i-th term in of R_n in (3) and $\otimes^2$ denotes the outer product of a vector with itself (i.e., $v^{\otimes^2} = vv^\top$). Standard errors and in turn confidence intervals for the individual parameters can thus be obtained by extracting the corresponding diagonal elements of $\widehat{\Omega}\widehat{M}\widehat{\Omega}^\top/n$. The empirical coverage of the confidence intervals obtained in this way is studied in §4.

2.2 | Optimization Algorithm

To (approximately) minimize the objective function (3), we combine the popular proximal gradient descent method (Beck and Teboulle 2009) with coordinate descent updates for the scale parameter ϕ. Specifically, we perform a fixed number of iterations of proximal gradient descent in (β_0, β) for fixed ϕ, and then update ϕ using one-dimensional minimization for fixed (β_0, β). We alternate between these two steps until convergence. A summary is provided in Algorithm 1, with specific details discussed below. We note that the objective function (3) is not globally convex, i.e., convergence to a global optimum cannot be guaranteed. However, under suitable condition (cf. §3), it can be shown that (3) is convex in a neighborhood of θ^*, i.e., assuming that the global minimizer and the initial iterate $\theta^{(0)} = (\beta_0^{(0)}, \beta^{(0)}, \phi^{(0)})$ are contained in that neighborhood, global optimality can be achieved.

Initial iterate. The following options are considered for obtaining $(\beta_0^{(0)}, \beta^{(0)})$: (i) "plain" Beta regression without penalty ($\lambda_n = 0$) using the betareg package (Cribari-Neto and Zeileis 2010), or (ii) ℓ_1-penalized logistic regression as implemented, e.g., in glmnet (Friedman et al. 2010), ignoring the fact the responses are not in $\{0,1\}$ and that the model is

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Algorithm 1 Optimization algorithm for minimization of (3)

Set $t \leftarrow 0$, initialize $\beta_0^{(0)}, \beta^{(0)}, \phi^{(0)}$. Determine stepsize $s > 0$ and fix an integer $M > 0$.

while true

$$\beta_0' \leftarrow \beta_0^{(t)} - s\nabla_{\beta_0} R_n(\theta^{(t)}), \quad \beta' \leftarrow \mathsf{S}_{\lambda \cdot s}(\beta^{(t)} - s\nabla_{\beta} R_n(\theta^{(t)})), \tag{6}$$

(here, $\mathsf{S}_\tau(z) := \left(\max\{|z_j| - \tau, 0\}\text{sign}(z_j)\right)_{j=1}^p$ denotes the soft shrinkage operator).

if condition (7) with $\theta' = (\beta_0', \beta')$ is not met, set $s \leftarrow .9 \cdot s$, and re-evaluate (6)

else

$t \leftarrow t + 1$; $\beta_0^{(t)} \leftarrow \beta_0'$; $\beta^{(t)} \leftarrow \beta'$

if $t \bmod M = 0$

$\phi^{(t)} = \arg\min_\phi R_n(\beta_0^{(t)}, \beta^{(t)}, \phi)$

else

$\phi^{(t)} = \phi^{(t-1)}$

end if

end if

if $R_n(\theta^{(t-1)}) - R_n(\theta^{(t)}) < \text{tol}$ (here, tol denotes a numerical tolerance).

break

end if

end while

misspecified. Regardless, this appears a reasonable approach particularly if (i) is not feasible (as is the case for $p > n$) because the associated negative log-likelihood function can still be interpreted as the cross-entropy loss (or "deviance") comparing observed and fitted proportions.

Given $(\beta_0^{(0)}, \beta^{(0)})$, we obtain $\phi^{(0)}$ via a coordinate descent step.

Choice of the stepsize. For the proximal gradient descent method with a fixed stepsize, the default choice in the globally convex case is given by $s = 1/L$, with L being equal to the Lipschitz constant of the gradient. Since the latter is often not easy to evaluate, it is common to use the upper bound $\sup_z \|\nabla^2 f(z)\|$, where $\nabla^2 f(z)$ denotes the Hessian (at some point z) of the objective f to be minimized and $\|\cdot\|$ here denotes the spectral norm (i.e., the maximum eigenvalue). Accordingly, we use $L = 1/\|\nabla^2 R_n(\theta^{(0)})\|$, where $\theta^{(0)}$ is the initial iterate. In case it is computationally prohibitive to materialize $\nabla^2 R_n(\theta^{(0)})$, we use that $\nabla^2 R_n(\theta^{(0)}) = \frac{1}{n}\mathbf{X}^\top W(\theta^{(0)})\mathbf{X}$ with $W = \text{diag}(w_1(\theta^{(0)}), \ldots, w_n(\theta^{(0)}))$ and the resulting upper bound $\|\nabla^2 R_n(\theta^{(0)})\| \le \max_{1 \le i \le n} w_i(\theta^{(0)})\|\mathbf{X}\|^2/n$, where $\mathbf{X}$ denotes the design matrix (including intercept) and the $\{w_i(\theta^{(0)})\}_{i=1}^n$ are observation-specific "weights".

A valid choice of stepsize ensures that the condition

$$R_n(\theta') \le R_n(\theta^{(t)}) + \langle \nabla_{\beta_0,\beta} R_n(\theta^{(t)}), \theta' - \theta^{(t)} \rangle + \frac{1}{2s}\|\theta' - \theta^{(t)}\|_2^2 \tag{7}$$

is satisfied for all updates from $\theta^{(t)}$ to a new iterate θ', where the right hand side is used as a (local) majorization of the objective R_n around $\theta^{(t)}$. If the condition (7) fails to hold, it is reduced by a factor less than one, and a new update is attempted until the condition is met. This procedure is in line with backtracking line search strategies employed for proximal gradient descent methods.

3 | Analysis

In this section, we state a non-asymptotic bound on the ℓ_1-estimation error in a neighborhood of the true parameter, thereby establishing high-dimensional consistency of our approach over such neighborhood. Note that the non-convexity of the underlying objective function renders the analysis much more challenging, and as a result, existing literature has adopted a localized analysis approach. Our analysis is based on the general strategy developed in (Elsener and van de Geer 2018).

For simplicity, we assume that the scale parameter ϕ^* is known. Moreover, we drop the intercept from the model, i.e., we assume that $\beta_0^* = 0$; alternatively, we may absorb the intercept in the X's. With these simplifications in place, we re-state the estimation problem of interest as follows.

$$\widehat{\beta} \in \arg\min_\beta \{R_n(\beta) + \lambda_n \|\beta\|_1\}, \tag{8}$$

In the sequel, we assume that the X_i's are random, and let $R(\beta) = \mathbf{E}[R_n(\beta)]$, $\beta \in \mathbb{R}^p$.

Assumptions.

(A1) The X_i's, $1 \le i \le n$, are i.i.d. copies of a sub-Gaussian random vector X with sub-Gaussian norm (Vershynin 2018, §2.5) $\sup_{v:\|v\|_2 \le 1} \|\langle X, v\rangle\|_{\psi_2} < C_X < \infty$.

(A2) There exists $\underline{\mu} > 0$ such that $\min_{v:\|v\|_2=1} \{v^\top \nabla^2 R(\beta^*) v\} \ge \underline{\mu}$.

(A3)

$$\mathbf{P}(|X^\top \beta| \le h) = 1 \text{ for all } \beta \in \mathcal{B}, \text{ where } \mathcal{B} = \{\beta: \|\beta - \beta^*\|_2 \le \eta\}$$

with radius $\eta > 0$ as defined in Lemma 1. In particular, with probability one, it holds that $0 < c_h \le \mu_i(\beta) \le 1 - c_h < 1$, $1 \le i \le n$.

The following lemma bounds the minimum eigenvalue of $\nabla^2 R(\beta)$ in a neighborhood of β^*, which yields strong convexity of R in such a neighborhood. It is an important intermediate step in establishing our main result via Theorem A.1 in the appendix.

Lemma 1. *Consider Assumption **(A3)**. There exists a constant $\overline{L}_{h,\phi^*}$ depending only on ϕ^* and h so that if $\eta = \overline{L}_{h,\phi^*}^{-1} K_X^{-1} \underline{\mu}/2$ with $K_X = 2^{3/2} C_X^3$ and C_X as in **(A1)**, it holds that*

$$\min_{\beta \in \mathcal{B}} \left\{ \min_{v:\|v\|_2=1} v^\top \nabla^2 R(\beta) v \right\} \ge \underline{\mu}/2.$$

A proof of Lemma 1 is provided in the appendix.

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The following statement contains the main result. It establishes a high probability, non-asymptotic bound on the estimation error in ℓ_1-norm, and in turn high-dimensional consistency.

Theorem 1. *Let* $\lambda_n > \lambda_\epsilon$ *with*

$$\lambda_\epsilon = \phi^* \left\{ 2\log(2n)\left[\sqrt{\frac{\log p}{n}} + \frac{8h}{3}\frac{\log p}{n}\right] + C_{h,\phi^*} C_X \sqrt{\frac{\log p}{n}} + \tilde{C}_{h,\phi^*}\frac{\log p}{n} \right\} + C_X \overline{C}_{h,\phi^*}\left(\sqrt{\frac{\log(p+1)}{n}} + \frac{\log(p+1)}{n}\right).$$

Under Assumptions ***(A1)***, ***(A2)***, *and* ***(A3)***, *for any stationary point* $\widehat{\beta}$ *of (8) contained in* $\mathcal{B}$ *it holds that*

$$\|\widehat{\beta} - \beta^*\|_1 \leq \frac{2}{\underline{\mu}} \frac{\{\frac{3}{2}\lambda + \frac{1}{2}\lambda_\epsilon\}^2}{\lambda - \lambda_\epsilon} s, \qquad s := \sum_{j=1}^{p} \mathbf{1}_{\beta_j^* \neq 0},$$

with probability at least $1 - C_{h,\phi^*}/n - \log_2(\eta n\sqrt{p})/p$, *where* $C_{h,\phi^*}, \tilde{C}_{h,\phi^*}$ *and* $\overline{C}_{h,\phi^*}$ *are positive constants depending only on* h *and* ϕ^*.

To better understand the implications of the above theorem, let us consider the choice $\lambda_n \asymp \lambda_\epsilon \asymp \log(n)\sqrt{\log(p)/n}$ under the mild assumption that $n \geq \log p$, in which case $\|\widehat{\beta} - \beta^*\|_1 = O(s\lambda_\epsilon) = O(s\log(n)\sqrt{\log(p)/n})$ with s being the sparsity level, i.e., the number of non-zero coefficients in β^*. This yields consistency under the condition $s^2\log^2(n)\log(p)/n \to 0$ as $n \to \infty$, and roughly aligns with existing results for other high-dimensional generalized linear models (van de Geer 2008; van de Geer et al. 2014; Wainwright 2019). A notable difference is the presence of the extra $\log^2(n)$ factor, which in the proof arises from the sub-exponential tails of log-Beta random variables. It remains unclear whether this term can be avoided. The specific constants appearing in the theorem and their significance are provided in the appendix, which contains a detailed proof of the result.

We would like to point out that in accordance with the framework in (Elsener and van de Geer 2018), Theorem 1 makes a statement about stationary points of the penalized negative log-likelihood R_n in a neighborhood of β^*. Note that in practice, a stationary point generated by the proximal gradient descent algorithm proposed in §2.2 is not guaranteed to be contained in this neighborhood. Establishing such a result is not attempted herein but we point out that analyses of this flavor for other ℓ_1-penalized problems with non-convex objective exist in the literature (e.g., Loh and Wainwright 2015; Song et al. 2020; Loh and Wainwright 2012).

4 | Empirical Results

In this section, we present the results of simulation studies to back up the theoretical analysis in §3 and demonstrate the utility of the approach through a case study from social justice research.

4.1 | Simulations

Setup. Our simulations studies are broadly divided into (i) a low-dimensional setup with $s \in \{2,5,10\}$ and $p \in \{20,50,100\}$, and (ii) a high-dimensional setup with $s \in \{5,10,20\}$ and $p \in \{200,500,800\}$. In both settings, we let $n \in \{500,1000\}$. Here, s denotes the size of the support of the target β^*, which is generated as follows.

$$\beta^*_{2j-1} = \sqrt{1/(j+1)},\ \beta^*_{2j} = -\sqrt{1/(j+1)},\ j = 1, \ldots, \lceil s/2 \rceil,\ \beta_j = 0\ \forall j \geq s+1, \tag{9}$$

Subsequently, β^* is normalized to have squared Euclidean norm equal to 13/6. This number is the squared Euclidean norm of the vector obtained according to (9) when setting s = 6. Furthermore, we set β_0^* and $\phi^* = 4$. The covariates $X_i = (X_{i1}, \ldots, X_{ip})^\top$ are i.i.d. Gaussian random vectors with zero mean and identity covariance, $1 \leq i \leq n$. Conditional on X_i, the response Y_i is drawn from the Beta distribution in Equation (1), $1 \leq i \leq n$.

Given the data $\{(X_i, Y_i)\}_{i=1}^n$, we consider the optimization problem (4) with the choice $\lambda = 0.2\sqrt{\log(p)/n}$, where the rate $\sqrt{\log(p)/n}$ is the default in ℓ_1-penalized regression problems and the specific factor 0.2 was determined by experimenting with values from the grid $\{0.1, 0.2, \ldots, 1\}$ in trial runs. The proximal gradient algorithm in §2.2 is used to compute $(\widehat{\beta}_0, \widehat{\beta}, \widehat{\phi})$, and confidence intervals (CIs) via de-biasing are obtained by computing $\widehat{\Omega}$ in (5) with the code (Javanmard et al. 2025) and the choice $\lambda_0 = 0.01\sqrt{\log(p)/n}$. CIs are only computed for the low-dimensional setup (i). In the high-dimensional setup, coverage tends to be below the nominal level for most (p,s,n)-configurations, which does not come as a surprise since correctness of the CIs is tied too more stringent conditions than consistent parameter estimation even for plain linear models with sub-Gaussian errors (Javanmard and Montanari 2014; van de Geer et al. 2014). Every (p,s,n) configuration is run with 1,000 independent replications.

Results. The results of our simulations are summarized in Figure 1 and Table 2. The figures show boxplots of the ℓ_1-estimation errors $\|\widehat{\beta} - \beta^*\|_1$ in dependence of the sparsity levels s for different combinations of (n,p) on a log-log scale. The empirical errors are compared to predictions according to the linear model

$$\log\|\widehat{\beta} - \beta^*\|_1 = \gamma_0 + \gamma_1 \log s + \gamma_2 \log n + \gamma_3 \log\log p. \tag{10}$$

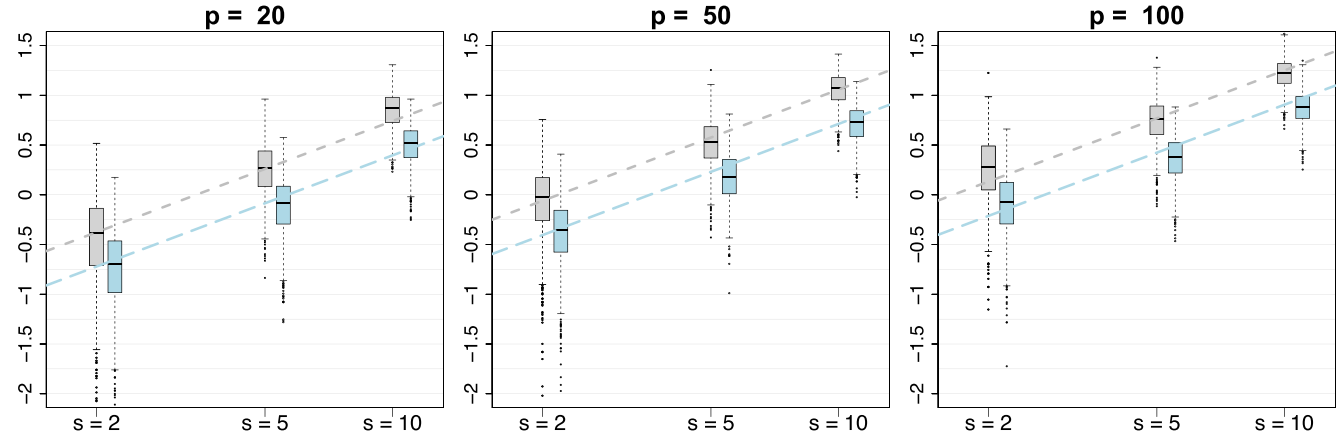
Based on Theorem 1 we expect that with a suitable choice of λ, we have $\gamma_1 = 1$, $\gamma = -1/2$, and $\gamma_3 \geq 1/2$. A linear regression model of the form (10) is fitted for both the low-dimensional and high-dimensional setup based on $3 \times 3 \times 2 \times 1{,}000 = 18{,}000$ simulation runs in each case. The above linear model achieves an R^2 of .77 and .85 in the low-dimensional and high-dimensional setup, respectively. The estimates for γ_2 are close to $-1/2$ in both cases as expected, while the estimates for γ_1 are below the expected value of 1. The estimates for γ_3 are around 1 and 2, respectively, confirming that the power in the polylog term in front of the $s/\sqrt{n}$ rate for the ℓ_1-estimation error is likely larger than 0.5, in alignment with Theorem 1.

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Table 2 contains average CI coverage rates (for the low-dimensional setup, in %) and true/false positive rates (TPRs and FPRs, in %) to evaluate the level of model parsimony achieved by means of ℓ_1-penalization, with true and false positives corresponding to the non-zero entries in $(\widehat{\beta}_j)_{j=1}^{s}$ and $(\widehat{\beta}_j)_{j=s+1}^{p}$, respectively. CI coverage rates are rather close to the nominal 95% for $n = 1{,}000$ and $p = 20$, which roughly corresponds to the regime encountered in the case study below. Coverage rates drop gradually (but not dramatically) as p and/or s are increased. For $n = 500$, the rates range from about 94% to 92.5% for $p \leq 50$ and $s \leq 5$, but to drop 90.2% for $s = 10$. As mentioned above, it is well-known that the success of the de-biasing approach for constructing CIs for ℓ_1-penalized estimators relies on the condition $s^2 \lesssim n$. Regarding TPRs and FPRs, we observe that the FPRs are consistently low and thus parsimonious model fits are generated. At the same time, the FPRs are also bounded away from zero, which means that selected models tend to include extraneous variables in addition to the support of β^*. This is well in line with the literature (Zhao and Yu 2006; Su et al. 2017). TPRs are at least in the nineties for $s \leq 10$. Note that $s = 20$, the coefficients in the support of β^* tend to be smaller in magnitude according to the protocol described after (9), increasing the chance of false non-selections due to the shrinkage induced by the ℓ_1-penalty.

4.2 | Case Study

Our case study uses data from (Ramezani et al. 2022) in which macro-county characteristics predictive of jail population size were identified for all 3,141 counties in the U.S. In this study, a beta regression model was fitted using a pre-selected set of variables to understand how demographic, socioeconomic, and variables pertaining to physical & mental health and the criminal legal system relate to jail population per capita (the target variable) at the county level. The incarceration trends database of the Vera institute (Vera Institute of Justice 2018a) was used to obtain incarceration counts. A variety of economic, social, demographic and public health-related variables were extracted from the Robert Wood Johnson Foundation's County Health Rankings and Roadmaps, which itself is compiled from several data sources provided by the University of Wisconsin (Vera Institute of Justice 2018b). Police data were



Regression model: $\log(\|\widehat{\beta} - \beta^*\|_1) = \gamma_0 + \gamma_1 \log s + \gamma_2 \log n + \gamma_3 \log\log p.$

	$\widehat{\gamma}_0$	$\widehat{\gamma}_1$	$\widehat{\gamma}_2$	$\widehat{\gamma}_3$	R^2
low-dim	.937 (.044)	.694 (.003)	−.498 (.006)	1.18 (.012)	.77
high-dim	−.270 (.034)	.502 (.002)	−.516 (.004).	2.21 (.013)	.85

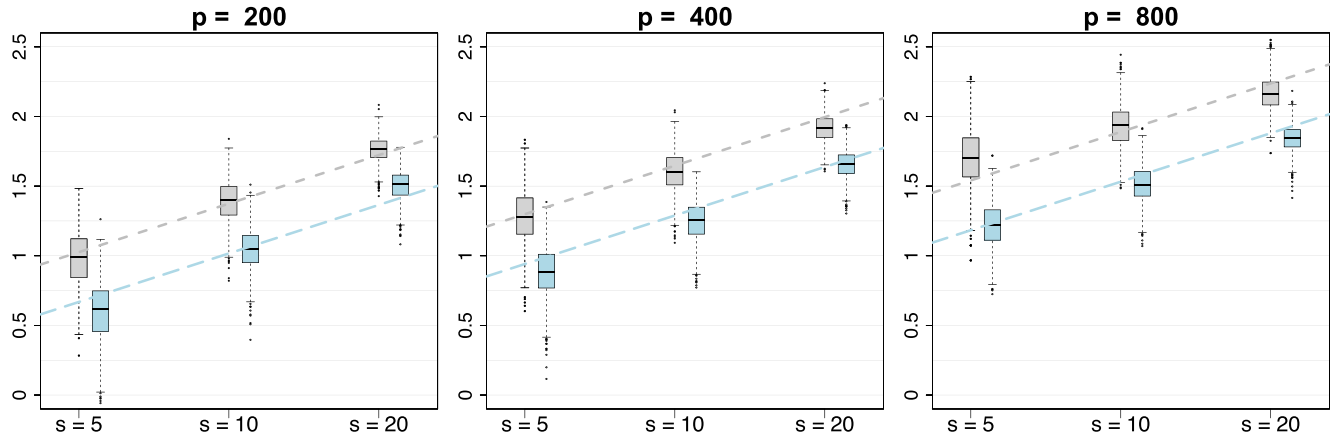


FIGURE 1 | Logarithmized ℓ_1-estimation error $\|\widehat{\beta} - \beta^*\|_1$ against s (axis marks on the horizontal axis also follow a log-scale), for different (n, p) combinations. Grey: $n = 500$; Blue: $n = 1{,}000$. The table in the center contains the estimated regression coefficients and the R^2s of the linear model (10) fitted for the low-dimensional (top) and the high-dimensional setup (bottom).

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TABLE 2 | Average coverage rates (CVG) and TPRs/FPRs (in %) with respect to the support of β^* for various configurations of (p, s, n).

		$n = 500$			$n = 1000$		
		p			p		
		20	**50**	**100**	**20**	**50**	**100**
	CVG	93.8 (.12)	92.8 (.17)	92.3 (.17)	95.1 (.06)	94.3 (.12)	93.5 (.14)
$s = 2$	TPR	100 (.00)	100 (.00)	100 (.00)	100 (.00)	100 (.00)	100 (.00)
	FPR	19.0 (.29)	13.9 (.16)	11.3 (.11)	18.5 (.30)	13.1 (.15)	10.4 (.10)
	CVG	93.7 (.11)	92.5 (.17)	91.7 (.22)	95.0 (.06)	93.7 (.12)	93.3 (.14)
$s = 5$	TPR	99.4 (.10)	99.1 (.13)	98.5 (.17)	100 (.00)	100 (.00)	100 (.00)
	FPR	19.3 (.32)	14.1 (.17)	11.4 (.11)	18.8 (.32)	13.5 (.17)	10.4 (.10)
	CVG	90.3 (.20)	90.2 (.24)	90.5 (.25)	94.7 (.06)	93.4 (.12)	93.2 (.14)
$s = 10$	TPR	88.6 (.31)	84.6 (.35)	81.2 (.37)	98.8 (.10)	98.0 (.14)	97.1 (.17)
	FPR	18.1 (.39)	14.2 (.18)	11.2 (.11)	18.7 (.40)	13.2 (.17)	10.3 (.10)
		$n = 500$			$n = 1000$		
		p			p		
		200	**400**	**800**	**200**	**400**	**800**
$s = 5$	TPR	97.8 (.21)	97.4 (.22)	97.1 (.23)	100 (.00)	100 (.03)	100 (.00)
	FPR	9.0 (.07)	7.4 (.05)	6.8 (.05)	8.2 (.07)	6.6 (.04)	5.5 (.03)
$s = 10$	TPR	78.0 (.39)	74.8 (.42)	71.7 (.42)	96.5 (.18)	95.4 (.21)	94.8 (.22)
	FPR	8.9 (.07)	7.3 (.05)	6.6 (.05)	8.2 (.06)	6.6 (.04)	5.5 (.03)
$s = 20$	TPR	52.5 (.32)	49.1 (.32)	46.1 (.32)	78.2 (.27)	74.9 (.29)	72.7 (.30)
	FPR	8.9 (.07)	7.2 (.05)	6.5 (.04)	8.1 (.07)	6.5 (.04)	5.4 (.03)

obtained from the Uniform Crime Report (Federal Bureau of Investigation U.S. Department of Justice 2011a, 2011b). The above data sources were linked using county and state identifiers, and per capita rates of the aforementioned variables were calculated using the population counts from the U.S. Census (Bureau 2020). The list of variables under consideration are tabulated in Table 3.

Results. The $p = 20$ variables are used as predictor variables for the target variable "proportion of individuals incarcerated". After eliminating counties for which any of the predictor variables is missing, we are left with $n = 2377$ observations. Prior to fitting our proposed ℓ_1-penalized Beta regression approach, all predictor variables are centered and scaled to unit empirical variance as customary in penalized regression. For ease of presentation, results pertaining to coefficient estimates reported below are with respect to the modified location and scales obtained through centering and scaling.

Figure 2 displays the trajectories of the coefficients $\{\widehat{\beta}_j(\lambda)\}_{j=1}^p$ for 50 values of λ spaced evenly (on a logarithmic scale) between 10^{-4} and $.95\overline{\lambda}$ with $\overline{\lambda} = \|\nabla R_n(\beta_0^\star, \mathbf{0}, \phi^\star)\|_\infty$ denoting the smallest value of λ for which $(\beta_0^\star, \mathbf{0}, \phi^\star)$ is a minimizer of the optimization problem (4) according to the KKT conditions; here, $\beta_0^\star$ and $\phi^\star$ represent the values of the MLE of these parameters in a plain intercept model.

TABLE 3 | List of predictor variables used in the case study.

Income inequality ratio	Community Mental Health Centers⋆
High School graduation rate	Violent crimes⋆
County size is medium (indicator)	Police officers⋆
County size is large (indicator)	Median household income
Poor physical health days	% African American
Poor mental health days	% Hispanic
Primary Care Physicians⋆	% Rural population
Healthcare costs⋆	% Over Age of 18
% drug treatment paid by Medicaid	% Children in poverty
Psychiatrists⋆	HPSA Score∗

Starred variables are per capita.
∗: Health Professional Shortage Area score.

For each value of λ, we run the proximal gradient method in §2.2 with stepsize $s = .5 \cdot \|\nabla^2_{\beta_0,\beta} R_n(\widehat{\beta}_0(0), \widehat{\beta}(0), \widehat{\phi}(0)\|$, where $(\widehat{\beta}_0(0), \widehat{\beta}(0), \widehat{\phi}(0))$ denote the (unpenalized) solution with $\lambda = 0$

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obtained through the betareg package, which also serves as initial solution. The scale parameter ϕ is updated every $M = 5$ iterations.

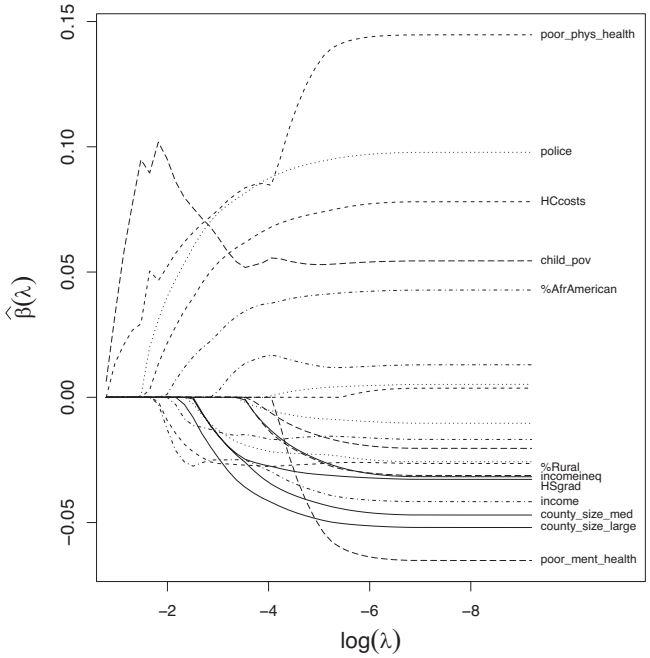


FIGURE 2 | Coefficient trajectories $\{\widehat{\beta}_j(\lambda)\}_{j=1}^{p}$ for the case study. Abbreviations of the variables are given at the right.

In addition, we compute the solution for $\lambda = 2\sqrt{\log(p)/n}$ that is roughly in line with the choice of the tuning parameter according to the theoretical analysis in §3. Table 4 lists the resulting coefficient estimates and their confidence intervals where the matrix $\widehat{\Omega}$ is computed with the implementation of (Javanmard and Montanari 2014) available from (Javanmard et al. 2025). As a reference, Table 4 also lists the coefficient estimates of the AIC-optimal model obtained via exhaustive enumeration of all $2^p \approx 10^6$ variable subsets.

Discussion. The AIC-optimal model produces a more parsimonious solution than ℓ_1-penalization with the above choice of the tuning parameter (8 vs. 12 non-zero coefficients). The latter does not miss any of the non-zero variables of the AIC-optimal model and the sets of prominent variables with comparatively large coefficients (Poor physical health days, Health Care costs, Police officers, and % Children in poverty) agree fully. As expected, the coefficients of the ℓ_1-penalization approach undergo significant shrinkage towards zero, which likely explains the over-selection of extraneous variables to compensate for this shrinkage (Zhao and Yu 2006). Observe that zero is contained in each confidence intervals for variables with a coefficient estimate of zero. At the same time, none of the confidence intervals of the variables included in the AIC-optimal contain zero; all except for one of the intervals (the one for "County size is large") include the coefficient estimate of the AIC-optimal model. Among the four extra variables included in the ℓ_1-penalization model, the confidence intervals of two of these ("Psychiatrists" and "% Population over the age of 18") contain zero; for, the variables "% African

TABLE 4 | Coefficient estimates $\{\widehat{\beta}_j\}$, their confidence intervals (note that by construction, these may not need to be centered at the parameter estimates), and the corresponding coefficient estimates $\{\widehat{\beta}_j^{\text{AIC}}\}$ in the AIC-optimal model. For space reasons, we here list the intercepts $\widehat{\beta}_0 = -5.28$ and $\widehat{\beta}_0^{\text{AIC}} = -5.29$ and scale parameters $\widehat{\phi} = 375.3$ and $\widehat{\phi}^{\text{AIC}} = 384.3$.

	Inc. ineq.	**HS grad. rate**	**Size=medium**	**Size=large**	**Poor Phys.**
$\widehat{\beta}_j$	0	−.0041	−.0048	−.0127	.0683
CI	[− .032, .032]	[− .053, − .001]	[− .065, − .015]	[− .043, − .007]	[.028, .173]
$\widehat{\beta}_j^{\text{AIC}}$	0	−.0397	−.0472	−.0725	.0764
	Poor Ment.	**PCP**	**HC cost**	**%Medicaid**	**Psychiatrists**
$\widehat{\beta}_j$	0	0	.0439	−.0234	−.0057
CI	[− .062, .062]	[− .033, .033]	[.024, .095]	[− .098, − .044]	[− .026, .055]
$\widehat{\beta}_j^{\text{AIC}}$	0	0	.0751	−.0470	0
	Comm. MHC	**Violent crimes**	**Police officers**	**Household Inc.**	**%African**
$\widehat{\beta}_j$	0	0	.0641	−.0265	.0187
CI	[− .015, .015]	[− .046, .046]	[.024, .121]	[− .245, − .156]	[.018, .097]
$\widehat{\beta}_j^{\text{AIC}}$	0	0	.0828	0	0
	%Hispanic	**%Rural**	**%Over18**	**%ChildrenPov**	**HPSA**
$\widehat{\beta}_j$	0	0	−.0117	.0727	0
CI	[− .034, .034]	[− .042, .042]	[− .040, .046]	[.073, .214]	[− .027, .027]
$\widehat{\beta}_j^{\text{AIC}}$	0	0	0	.0968	0

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American“ and “Median Household income” the zero is not included, which constitutes the only noticeable discrepancy from the AIC-optimal model.

5 | Conclusion

In this paper, we have studied ℓ_1-penalized Beta regression in a high-dimensional setup. We have provided a non-asymptotic bound on the ℓ_1-estimation error of stationary points in a neighborhood of the true parameter, following the framework in (Elsener and van de Geer 2018) for ℓ_1-penalized M-estimators with non-convex objective function. To enable inference for low-dimensional parameters such as individual regression coefficients, we adopt the debiasing approach in (Zhang and Zhang 2014; van de Geer et al. 2014; Javanmard and Montanari 2014) that is widely used for ℓ_1-penalized estimators. We also present a scalable framework for optimization based on proximal gradient descent, a popular first-order algorithm. We apply our approach to select variables in a Beta regression model for predicting the fraction of incarcerated individuals in US counties. This case study demonstrates the utility of our approach.

Looking ahead, there are various avenues for future research. First, it is worth examining the statistical properties of stationary points returned by the proposed proximal gradient algorithm, thereby unifying computational and theoretical aspects. Second, in this paper we have focused on the ℓ_1-estimation error; studying the ℓ_2-estimation error and prediction error (with respect to the loss induced by the negative log-likelihood) will complement the results in this paper. Third, it is worth considering concave penalties such as the SCAD (Fan and Li 2001) and MCP (Zhang 2010) that reduce the bias affecting the estimation of the non-zero coefficients of the target. Fourth, in this paper we have focused on the homoscedastic Beta regression model with a constant scale parameter. In heteroscedastic Beta regression, observation-specific scale parameters may depend on covariates. The paper (Zhao et al. 2014) considers sparse estimation in this setting when the mean and scale are modeled by two distinct linear predictors. It is worth investigating this scenario in a high-dimensional setup.

Author Contributions

Niloofar Ramezani and Martin Slawski jointly contributed to the development of the methodological framework, theoretical analysis, and manuscript preparation. The study was initiated and primarily led by Niloofar Ramezani, who implemented the computational algorithms and coordinated the simulation and data analysis. Martin Slawski provided expertise in optimization and theoretical developments, offering critical input throughout the methodological development. Both authors collaborated on interpreting results, revising the manuscript, and approved the final version.

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Appendix A

Proof of Lemma 1. *Let $v \in \mathbb{R}^p$ be arbitrary. Invoking Lemma A.1, we have*

$$v^\top(\nabla^2 R(\beta) - \nabla^2 R(\beta^*))v = \mathbf{E}[(X^\top v)^2\{\rho_X(X^\top\beta) - \rho_X(X^\top\beta^*)\}], \tag{A1}$$

where the map $\rho\colon \mathbb{R} \to \mathbb{R}$ is given by

$$\begin{aligned}\rho_X(z) = &\{\mu'(z)\}^2\phi^{*2}\{\Psi'(\mu(z)\phi^*) + \Psi'((1-\mu(z))\phi^*)\}\\ &+\mu''(z)\phi^*\{[\Psi(\mu(z)\phi^*) - \Psi(\mu(X^\top\beta^*)\phi^*)] -\\ &[\Psi((1-\mu(z))\phi^*) - \Psi((1-\mu(X^\top\beta^*))\phi^*)]\},\ z\in\mathbb{R}.\end{aligned}$$

In the sequel, we will show that ρ_X is Lipschitz on $[-h,h]$. For this purpose, we let L_f denote the Lipschitz constant of some function f. Using the properties established in Lemmas A.3 and A.4, we obtain that

$$\begin{aligned}L_{\rho_X} &\le 2\phi^{*2}L_{\mu'}\|\mu'\|_\infty 2L_\Psi + \phi^{*2}\|(\mu')^2\|_\infty\{2L_{\Psi'}L_\mu\phi^*\}\\ &\quad + 4\phi^* L_{\mu''}\|\Psi\|_\infty + 2L_{\Psi'}\phi^*\|\mu''\|_\infty\\ &\le \frac{1}{4}\phi^{*2}C'_{h,\phi} + \frac{1}{32}\phi^{*3}C''_{h,\phi} + \frac{3}{2}\phi^* C_{h,\phi} + \frac{1}{2}\phi^* C''_{h,\phi} =: \overline{L}_{h,\phi^*}.\end{aligned}$$

Consequently, in light of (A1),

$$\begin{aligned}v^\top(\nabla^2 R(\beta) - \nabla^2 R(\beta^*))v &\le \overline{L}_{h,\phi^*}\mathbf{E}[(X^\top v)^2|X^\top(\beta-\beta^*)|]\\ &\le \overline{L}_{h,\phi^*}\sqrt{\mathbf{E}[(X^\top v)^4]}\sqrt{\mathbf{E}[\{X^\top(\beta-\beta^*)\}^2]}\\ &\le \overline{L}_{h,\phi^*}\|v\|_2^2 2^{3/2}C_X^3\|\beta-\beta^*\|_2,\end{aligned}$$

using that $\mathbf{E}[(X^\top v/\|v\|)^4] \le 4C_X^4$ and $\mathbf{E}[\{X^\top(\beta-\beta^*)/\|\beta-\beta^*\|_2\}^2] \le 2C_X^2$ according to well-known bounds on the moments of ψ_2-random variables. The assertion of the lemma then follows from $v^\top R(\beta^*)v \ge \underline{\mu}\|v\|_2^2$.

Proof of Theorem 1. *The result of the Theorem is obtained verifying the conditions of Theorem A.1. We first note that by assumption and according to Lemma 1, the risk R is $\underline{\mu}/2$- strongly convex in $\mathcal{B} = \{\beta\in\mathbb{R}^p\colon \|\beta-\beta^*\|_2 \le \eta\}$. The rest of the proof is devoted towards demonstrating that*

$$\sup_{\beta\in\mathcal{B}}\left|\{\nabla R_n(\beta) - \nabla R(\beta)\}^\top(\beta-\beta^*)\right| \le \lambda_\epsilon\|\beta-\beta^*\|_1 \tag{A2}$$

with high probability (to be specified), for a suitable choice of $\lambda_\epsilon > 0$, which will prompt the assertion of the theorem in virtue of Theorem A.1. Establishing (A2) relies on empirical process techniques, following the basic template developed in (Elsener and van de Geer 2018). That template is combined with an additional truncation step to cope with the unboundedness of the terms $\{\log(Y_i), \log(1-Y_i)\}_{i=1}^n$ appearing in ∇R_n. Below, we list the main steps to obtain (A2).

1. Establish (A2) for $\mathcal{B}_M = \{\beta\in\mathcal{B}\colon \|\beta-\beta^*\|_1 \le M\}$ using the concentration inequality in Lemma A.6 combined with a truncation step.
2. Bound the individual terms appearing in Lemma A.6 for the specific scenario under consideration here.
3. Extend the result from $\mathcal{B}_M$ to $\mathcal{B}$ via the peeling device (van de Geer 2000).

Step 1.

Define the class of functions

$$\begin{aligned}&\{f_\beta(x,y) := \phi^*\cdot x^\top(\beta-\beta^*)\mu'(x^\top\beta)\\ &\times\{[\Psi(\mu(x^\top\beta)\phi^*) - \log(y)] - [\Psi((1-\mu(x^\top\beta))\phi^*) - \log(1-y)]\},\\ &(x,y)\in\mathbb{R}^p\times(0,1)\}_{\beta\in\mathbb{R}^p}.\end{aligned}$$

Accordingly, we have

$$\begin{aligned}&\nabla R_n(\beta)^\top(\beta-\beta^*) = \frac{1}{n}\sum_{i=1}^n f_\beta(X_i,Y_i)\\ &\{\nabla R_n(\beta) - \nabla R(\beta)\}^\top(\beta-\beta^*) = \frac{1}{n}\sum_{i=1}^n\{f_\beta(X_i,Y_i) - \mathbf{E}[f_\beta(X_i,Y_i)]\}.\end{aligned}$$

Let further

$$\mathbf{Z}_M = \sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n\{f_\beta(X_i,Y_i) - \mathbf{E}[f_\beta(X_i,Y_i)]\}\right|.$$

For $y\in(0,1)$, define the map $m(y) = \max\{\log(y), \log(1-y)\}$, and let $\{t_M\}$ and u be arbitrary positive numbers. We then have

$$\begin{aligned}&\mathbf{P}(\exists M\colon \mathbf{Z}_M > t_M)\\ &= \mathbf{P}\left(\exists M\colon \sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n\{f_\beta(X_i,Y_i) - \mathbf{E}[f_\beta(X_i,Y_i)]\}\right| > t_M\right)\\ &\le \mathbf{P}\left(\exists M\colon \sup_{\beta\in\mathcal{B}_M}\left\{\left|\frac{1}{n}\sum_{i=1}^n\{f_\beta(X_i,Y_i)\mathbf{1}_{\{m(Y_i)\le u\}} - \mathbf{E}[f_\beta(X_i,Y_i)]\}\right|\right.\right.\\ &\quad\left.\left. + \left|\frac{1}{n}\sum_{i=1}^n f_\beta(X_i,Y_i)\mathbf{1}_{\{m(Y_i)>u\}}\right| > t_M\right\}\right)\\ &\le \mathbf{P}\left(\exists M\colon \sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n\{f_\beta(X_i,Y_i)\mathbf{1}_{\{m(Y_i)\le u\}} - \mathbf{E}[f_\beta(X_i,Y_i)]\}\right|\right.\\ &\quad\left. + \sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n f_\beta(X_i,Y_i)\mathbf{1}_{\{m(Y_i)>u\}}\right| > t_M\right)\\ &\le \mathbf{P}\left(\exists M\colon \sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n\{f_\beta(X_i,Y_i)\mathbf{1}_{\{m(Y_i)\le u\}} - \mathbf{E}[f_\beta(X_i,Y_i)]\}\right| > t_M\right)\\ &\quad + \mathbf{P}\left(\exists M\colon \sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n f_\beta(X_i,Y_i)\mathbf{1}_{\{m(Y_i)>u\}}\right| > 0\right)\end{aligned} \tag{A3}$$

Regarding the second term on the right hand side of the last inequality, we use the bound

$$\begin{aligned}&\mathbf{P}\left(\exists M\colon \sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n f_\beta(X_i,Y_i)\mathbf{1}_{\{m(Y_i)>u\}}\right| > 0\right)\\ &= 1 - \left(\forall M\colon \sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n f_\beta(X_i,Y_i)\mathbf{1}_{\{m(Y_i)>u\}}\right| = 0\right)\\ &\le 1 - \left(\bigcap_{i=1}^n \mathbf{1}_{\{m(Y_i)\le u\}}\right) \le C_{\phi^*,h}\frac{1}{n^{\kappa-1}},\end{aligned} \tag{A4}$$

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for some constant $C_{\phi^*,h} > 0$ depending only on ϕ^* and h (cf. Assumption **(A3)**) and $\kappa > 0$ to be chosen below, where the last inequality in (A4) is obtained by invoking Lemma A.2 with the choice

$$\begin{aligned} t \quad &= u = \frac{\kappa \log 2n}{\phi^* \min\limits_{1\le i\le n} \min\{\mu_i(\beta^*), 1-\mu_i(\beta^*)\}} \\ &\le \frac{\kappa \log 2n}{c_h \phi^*} =: \tilde{C}_{h,\phi^*}\, \kappa \log 2n \end{aligned} \tag{A5}$$

and c_h as defined in Assumption **(A3)**, and then using the union bound over $1 \le i \le n$.

Regarding the first term on the right hand side in (A3), we have

$$\begin{aligned} &\mathbf{P}\left(\beta\mathcal{B}_M \in \sup\left| \frac{1}{n}\sum_{i=1}^n \{f_\beta(X_i,Y_i)\mathbf{1}_{\{m(Y_i)\le u\}} - \mathbf{E}[f_\beta(X_i,Y_i)]\}\right| > t_M \right) \\ &= \mathbf{P}\Bigg(\sup_{\beta\in\mathcal{B}_M}\Bigg| \frac{1}{n}\sum_{i=1}^n \{f_\beta(X_i,Y_i)\mathbf{1}_{\{m(Y_i)\le u\}} - \mathbf{E}[f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)\le u}] \\ &\quad - \mathbf{E}[f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)>u}]\}\Bigg| > t_M \Bigg) \\ &\le \mathbf{P}\Bigg(\underbrace{\sup_{\beta\in\mathcal{B}_M}\left| \frac{1}{n}\sum_{i=1}^n \{f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)\le u} - \mathbf{E}[f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)\le u}]\}\right|}_{(I)} \\ &\quad + \underbrace{\beta\mathcal{B}_M \in \sup\left| \frac{1}{n}\sum_{i=1}^n \mathbf{E}[f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)>u}]\right| > t_M}_{(II)} \Bigg). \end{aligned} \tag{A6}$$

Now observe that for (II), we have

$$\begin{aligned} \sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n \mathbf{E}[f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)>u}]\right| &= \sup_{\beta\in\mathcal{B}_M}\left|\mathbf{E}[f_\beta(X,Y)\mathbf{1}_{m(Y)>u}]\right| \\ &\le \sup_{\beta\in\mathcal{B}_M}\sqrt{\mathbf{E}[f_\beta^2(X,Y)]}\sqrt{\mathbf{E}[\mathbf{1}_{m(Y)>u}]} \\ &\le n^{-\kappa/2}\sup_{\beta\in\mathcal{B}_M}\sqrt{\mathbf{E}[f_\beta^2(X,Y)]} \end{aligned} \tag{A7}$$

by choosing u as in (A5). Furthermore, we have for any $\beta \in \mathcal{B}_M$

$$\begin{aligned} \mathbf{E}[f_\beta^2(X,Y)] \quad &\le \frac{\phi^{*2}}{4}\mathbf{E}\left[\{X^\top(\beta-\beta^*)\}^2(4\log(Y)^2 + 4\log(1-Y)^2)\right] + \\ &\quad + \frac{\phi^{*2}}{4}\overline{C}_{h,\phi^*}\mathbf{E}[\{X^\top(\beta-\beta^*)\}^2] \end{aligned} \tag{A8}$$

where $\overline{C}_{h,\phi^*} = 4\,\|\Psi\|_\infty = 4C_{h,\phi^*}$ per Lemma A.4. Furthermore, we have

$$\begin{aligned} &\mathbf{E}\left[\{X^\top(\beta-\beta^*)\}^2(4\log(Y)^2 + 4\log(1-Y)^2)\right] \\ &= \mathbf{E}\left[\{X^\top(\beta-\beta^*)\}^2\mathbf{E}\left[(4\log(Y)^2 + 4\log(1-Y)^2)|X\right]\right] \\ &\le \check{C}_{\phi^*,h}\mathbf{E}\left[\{X^\top(\beta-\beta^*)\}^2\right], \end{aligned} \tag{A9}$$

for some constant $\check{C}_{\phi^*,h} > 0$ depending only on ϕ^* and h, using the moment bounds for log-beta random variables in Lemma A.2. We then use that for all $\beta \in \mathcal{B}_M$, we have

$$\mathbf{E}[X^\top(\beta-\beta^*)\}^2] \le 2C_X \cdot M^2, \tag{A10}$$

where $C_X = \|X\|_{\psi_2}$. Combining (A7) through (A10), we obtain that

$$\sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n \mathbf{E}[f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)>u}]\right| \le n^{-\kappa/2}\overline{C}_{\phi^*,h}\cdot C_X \cdot M, \tag{A11}$$

for $\kappa > 0$ to be chosen below.

Step 2.

We now turn our attention to term (I) in (A6), i.e., the supremum of an empirical process, to be targeted with Lemma A.6. We start by controlling the expectation of the supremum of the empirical process using Rademacher symmetrization (e.g., Wainwright 2019) and the contraction principle in the form of Lemma A.5. Let $\{\epsilon_i\}_{i=1}^n$ be an i.i.d. sequence of Rademacher random variables independent of $\{(X_i,Y_i)\}_{i=1}^n$. We then obtain

$$\begin{aligned} &\mathbf{E}\left[\sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n \{f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)\le u} - \mathbf{E}[f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)\le u}]\right|\right] \\ &\le 2\mathbf{E}_{\{(X_i,Y_i,\epsilon_i)\}}\left[\sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n \epsilon_i f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)\le u}\right|\right] \end{aligned} \tag{A12}$$

To apply the contraction principle in Lemma A.5, we condition on the $\{(X_i,Y_i)\}_{i=1}^n$ and then apply the Lemma with (i) $z_i = X_i$, $1 \le i \le n$, (ii) the class of functions $\mathcal{G} = \{z \mapsto g_\beta(z) = z^\top\beta,\ \beta \in \mathcal{B}_M\}$, (iii) the functions $\gamma_i : [-h, h] \to \mathbb{R}$ defined by

$$\begin{aligned} t \mapsto \gamma_i(t) = \big(&\phi^* t\mu'(t)\{[\Psi(\mu(t)\phi^*) - \log(Y_i)] \\ &- [\Psi((1-\mu(t))\phi^*) - \log(1-Y_i)]\} \\ &- \phi^* X_i^\top\beta^*\mu'(t)\{[\Psi(\mu(t)\phi^*) - \log(Y_i)] \\ &- [\Psi((1-\mu(t))\phi^*) - \log(1-Y_i)]\}\big)\mathbf{1}_{m(Y_i)\le u}, \end{aligned}$$

$1 \le i \le n$, and (iv) $g^* = g_{\beta^*}$ (observe that $\gamma_i(g_{\beta^*}(z_i)) = 0$, $1 \le i \le n$). Accordingly,

$$\begin{aligned} &\mathbf{E}_{\{(X_i,Y_i,\epsilon_i)\}}\left[\sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n \epsilon_i f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)\le u}\right|\right] \\ &= \mathbf{E}_{\{(X_i,Y_i)\}}\left[\mathbf{E}_{\{\epsilon_i\}_{i=1}^n}\left[\sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n \epsilon_i f_\beta(X_i,Y_i)\mathbf{1}_{m(Y_i)\le u}\right|\,\middle|\,\{(X_i,Y_i)\}_{i=1}^n\right]\right] \\ &= \mathbf{E}_{\{(X_i,Y_i)\}}\left[\mathbf{E}_{\{\epsilon_i\}_{i=1}^n}\left[\sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n \epsilon_i \gamma_i(g_\beta(X_i))\right|\,\middle|\,\{(X_i,Y_i)\}_{i=1}^n\right]\right] \\ &= \mathbf{E}_{\{(X_i,Y_i)\}} \\ &\quad\left[\mathbf{E}_{\{\epsilon_i\}_{i=1}^n}\left[\sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n \epsilon_i\{\gamma_i(g_\beta(X_i)) - \gamma_i(g^*(X_i))\}\right|\,\middle|\,\{(X_i,Y_i)\}_{i=1}^n\right]\right] \\ &\le 2\mathbf{E}_{\{(X_i,Y_i)\}}\left[\max_{1\le i\le n} L_{\gamma_i}\mathbf{E}_{\{\epsilon_i\}_{i=1}^n}\right. \\ &\quad\left.\left[\sup_{\beta\in\mathcal{B}_M}\left|\frac{1}{n}\sum_{i=1}^n \epsilon_i\{g_\beta(X_i) - g^*(X_i)\}\right|\,\middle|\,\{(X_i,Y_i)\}_{i=1}^n\right]\right], \end{aligned} \tag{A13}$$

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where the inequality is obtained from Lemma A.5, with L_{γ_i} denoting the Lipschitz constant of γ_i, $1 \le i \le n$. In the sequel, we will bound these Lipschitz constants. First note that if $m(Y_i) > u$, then $\gamma_i \equiv 0$, $1 \le i \le n$ and there is nothing to show. In the opposite case, we differentiate $t \mapsto \gamma_i(t)$, and bound the resulting derivative $t \mapsto \gamma_i'(t)$ on $[-h, h]$. For $1 \le i \le n$, letting $c_i = X_i^\top \beta^*$, we have

$$
\begin{aligned}
\gamma_i'(t) \quad &= \phi^* \underbrace{\mu'(t)\Psi(\mu(t)\phi^*)}_{\le \frac{1}{4} C_{h,\phi^*}} + \phi^* \underbrace{t\mu''(t)\Psi(\mu(t)\phi^*)}_{\le \frac{h}{4} C_{h,\phi^*}} + \phi^{*2} \underbrace{t\Psi'(\phi^*\mu(t))\{\mu'(t)\}^2}_{\le \frac{h}{16} C'_{h,\phi^*}} \\
&\quad - \phi^* \underbrace{\log(Y_i)\mu'(t)}_{\le u/4} - \phi^* \underbrace{t\log(Y_i)\mu''(t)}_{\le \frac{h\cdot u}{4}} \\
&\quad - \phi^* \underbrace{c_i \Psi(\mu(t)\phi^*)\mu''(t)}_{\le \frac{h}{4} C_{h,\phi^*}} - \phi^{*2} \underbrace{c_i \Psi'(\mu(t)\phi^*)\{\mu'(t)\}^2}_{\le \frac{1}{16} C'_{h,\phi^*}} \\
&\quad + \phi^* \underbrace{c_i \mu''(t)\log(Y_i)}_{\le \frac{h\cdot u}{4}} \\
&\quad - \phi^* \underbrace{\mu'(t)\Psi((1-\mu(t))\phi^*)}_{\le C_{h,\phi^*}/4} - \phi^* \underbrace{t\mu''(t)\Psi((1-\mu(t)\phi^*))}_{\le \frac{h}{4} C_{h,\phi^*}} \\
&\quad + \phi^{*2} \underbrace{t\Psi'((1-\mu(t))\phi^*)\{\mu'(t)\}^2}_{\le \frac{h}{16} C'_{h,\phi}} \\
&\quad + \phi^* \underbrace{\mu'(t)\log(1-Y_i)}_{\le \frac{u}{4}} + \phi^* \underbrace{t\mu''(t)\log(1-Y_i)}_{\le \frac{h\cdot u}{4}} \\
&\quad + \phi^* \underbrace{c_i \mu''(t)\Psi(\phi^*(1-\mu(t)))}_{\le \frac{h}{4} C_{h,\phi^*}} - \phi^{*2} \underbrace{c_i \Psi'((1-\mu(t))\phi^*)\{\mu'(t)\}^2}_{\le \frac{h}{16} C'_{h,\phi^*}} \\
&\quad - \phi^* \underbrace{c_i \mu''(t)\log(1-Y_i)}_{\le \frac{h\cdot u}{4}}, \quad t \in [-h, h]. \\
&\le u(h+1/2) + C_{h,\phi^*}(h+1/2) + hC'_{h,\phi^*}\phi^*/4 \le K_{h,\phi^*}\kappa \log 2n.
\end{aligned}
\tag{A14}
$$

where in the above display, all inequalities (which follow from Assumption **(A3)** and Lemma A.4) apply to the terms in absolute value. In the last inequality, we have inserted the choice for u in (A5) and $K_{h,\phi^*} > 0$ is a constant depending only on h, ϕ^*.

Combining (A12), (A13), and (A14), we obtain

$$
\begin{aligned}
&\mathbf{E}\left[\sup_{\beta \in \mathcal{B}_M} \left| \frac{1}{n} \sum_{i=1}^n \{f_\beta(X_i, Y_i)\mathbf{1}_{m(Y_i)\le u} - \mathbf{E}[f_\beta(X_i, Y_i)\mathbf{1}_{m(Y_i)\le u}] \right| \right] \\
&\le 4K_{h,\phi^*}\kappa \log(2n) \mathbf{E}_{\{(X_i,Y_i)\}} \\
&\quad \left[\mathbf{E}_{\{\epsilon_i\}} \left[\sup_{\beta \in \mathcal{B}_M} \left| \frac{1}{n} \sum_{i=1}^n \epsilon_i X_i^\top (\beta - \beta^*) \right| \,\middle|\, \{(X_i, Y_i)\}_{i=1}^n \right] \right] \\
&\le 4K_{h,\phi^*}\kappa \log(2n) \times M \cdot \mathbf{E}\left[\| \frac{1}{n} \sum_{i=1}^n \epsilon_i X_i \|_\infty \right] \\
&\le 4K_{h,\phi^*}\kappa \log(2n) \times M \cdot \left\{ \|(\epsilon_i)\|_{\psi_2} \|(X_{ij})\|_{\psi_2} \frac{\log(p+1)}{n} \right. \\
&\quad \left. + \sqrt{\frac{4\|(\epsilon_i)\|_{\psi_2}^2 \|(X_{ij})\|_{\psi_2}^2 \log(p+1)}{n}} \right\} \\
&=: M\tilde{\lambda}_1,
\end{aligned}
\tag{A15}
$$

where the last inequality can be taken directly from (Elsener and van de Geer 2018, cf. Appendix D.5).

To apply Bousquet's concentration inequality (Lemma A.6) to control the term (I) in (A6), it remains to determine $R > 0$ such that $\mathbf{E}[\{f_\beta(X_1, Y_1)\mathbf{1}_{m(Y_1)\le u} - \mathbf{E}[f_\beta(X_1, Y_1)\mathbf{1}_{m(Y_1)\le u}]\}^2] \le R^2$ and $K > 0$ such that $|\{f_\beta(X_1, Y_1)\mathbf{1}_{m(Y_1)\le u} - \mathbf{E}[f_\beta(X_1, Y_1)\mathbf{1}_{m(Y_1)\le u}]\}| \le K$ almost surely for all $\beta \in \mathcal{B}_M$. Regarding the former, we have

$$
\begin{aligned}
&\mathbf{E}[\{f_\beta(X_1, Y_1)\mathbf{1}_{m(Y_1)\le u} - \mathbf{E}[f_\beta(X_1, Y_1)\mathbf{1}_{m(Y_1)\le u}]\}^2] \\
&\le \mathbf{E}[f_\beta^2(X_1, Y_1)\mathbf{1}_{m(Y_1)\le u}] \\
&\le \mathbf{E}\Big[\phi^{*2}\{X_1^\top(\beta - \beta^*)\}^2 \{\mu'(X_1^\top\beta)\}^2 \big\{[\Psi(\mu(X_1^\top\beta)\phi^*) - \log(Y_1)] \\
&\qquad - [\Psi((1-\mu(X_1^\top\beta))\phi^*) - \log(1-Y_1)]\big\}^2 \mathbf{1}_{m(Y_1)\le u}\Big] \\
&\le \mathbf{E}\Big[\phi^{*2}\{X_1^\top(\beta - \beta^*)\}^2 \{\mu'(X_1^\top\beta)\}^2 \\
&\qquad \times 4\big\{[\Psi^2(\mu(X_1^\top\beta)\phi^*) + \log^2(Y_1)] \\
&\qquad + [\Psi((1-\mu(X_1^\top\beta))\phi^*)^2 + \log^2(1-Y_1)]\big\} \mathbf{1}_{m(Y_1)\le u}\Big] \\
&\le \frac{4}{16}\phi^{*2}[2\kappa^2 \log^2(2n) + 2(C'_{h,\phi^*})^2] \mathbf{E}[\{X_1^\top(\beta - \beta^*)\}^2] \\
&= \frac{4}{16}\phi^{*2}[2\kappa^2 \log^2(2n) + 2(C'_{h,\phi^*})^2] \mathbf{E}\left[\left\{ X_1^\top \frac{(\beta - \beta^*)}{\|\beta - \beta^*\|_2} \right\}^2 \right] \|\beta - \beta^*\|_2^2 \\
&\le M^2\phi^{*2}[\kappa^2 \log^2(2n) + (C'_{h,\phi^*})^2] \underbrace{\|X\|_{\psi_2}^2}_{\le C_X^2} =: R^2,
\end{aligned}
\tag{A16}
$$

where we have inserted the choice for u as above, and we have invoked Lemma A.4 again. Similarly, we find that almost surely

$$
\begin{aligned}
|\{f_\beta(X_1, Y_1)\mathbf{1}_{m(Y_1)\le u} - \mathbf{E}[f_\beta(X_1, Y_1)\mathbf{1}_{m(Y_1)\le u}]\}| \quad &\le 2h\phi^*\{C'_{h,\phi^*} + u\} \\
&\le 2h\phi^*\{C'_{h,\phi^*} + \kappa \log(2n)\}
\end{aligned}
\tag{A17}
$$

Combining (A15), (A16), (A17), (A11) and (A6), an application of Lemma A.6 with the choice

$$
\begin{aligned}
t_M = \quad & M\left\{ 2\tilde{\lambda}_1 + \phi^*[\kappa \log(2n) + C'_{h,\phi^*} C_X] \sqrt{\frac{t}{n}} + n^{-\kappa/2} \overline{C}_{\phi^*,h} C_X \right\} \\
& + \frac{8h\phi^*(C'_{h,\phi^*} + \kappa \log(2n))}{3} \frac{t}{n}
\end{aligned}
$$

for any $t > 0$ yields

$$
\mathbf{P}\left(\sup_{\beta \in \mathcal{B}_M} \underbrace{\left| \frac{1}{n} \sum_{i=1}^n \{f_\beta(X_i, Y_i)\mathbf{1}_{\{m(Y_i)\le u\}} - \mathbf{E}[f_\beta(X_i, Y_i)]\} \right|}_{=: \mathbf{Z}(\beta)} > t_M \right) \le \exp(-t).
\tag{A18}
$$

Defining

$$
\tilde{\lambda}_2(t) = \left\{ 2\tilde{\lambda}_1 + \phi^*[\kappa \log(2n) + C'_{h,\phi^*} C_X] \sqrt{\frac{t}{n}} + n^{-\kappa/2} \overline{C}_{\phi^*,h} C_X \right\}
$$

we can express (A18) as

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$$\mathbf{P}\left(\sup_{\beta\in B_M}\mathbf{Z}(\beta)\geq M\tilde{\lambda}_2(t)+\frac{8h\phi^*(C'_{h,\phi^*}+\kappa\log(2n))}{3}\frac{t}{n}\right)\leq\exp(-t). \tag{A19}$$

Step 3.

Note that the tail bound (A19) concerns a fixed value of M. The goal of this step is to obtain a tail bound for any "valid" M to extend the analysis from the set $\mathcal{B}_M$ to the set of interest $\mathcal{B}=\{\beta\in\mathbb{R}^p:\ \|\beta-\beta^*\|_2\leq\eta\}$. Observe that $\sup_{\beta\in\mathcal{B}}\|\beta-\beta^*\|_1\leq\eta\sqrt{p}$, thus it suffices to consider all $M\in[0,\eta\sqrt{p}]$. Consider the "slices"

$$\begin{aligned}S_0&=\left\{\beta\in\mathcal{B}\colon\|\beta-\beta^*\|_1\leq\frac{1}{n}\right\},\\ S_j&=\left\{\beta\in\mathcal{B}\colon\frac{2^{j-1}}{n}<\|\beta-\beta^*\|_1\leq\frac{2^j}{n}\right\},\\ &\quad j=1,\ \ldots,\lceil\log_2(\eta n\sqrt{p})\rceil,\end{aligned}$$

and observe that $\mathcal{B}\subseteq\bigcup_{j=0}^{\lceil\log_2(\eta n\sqrt{p})\rceil}S_j$. We start with the slice S_0. In view of (A19), we have

$$\mathbf{P}\left(\exists\beta\in S_0\colon\ \mathbf{Z}(\beta)\geq\frac{\tilde{\lambda}_2(t)}{n}+\frac{8h\phi^*(C'_{h,\phi^*}+\kappa\log(2n))}{3}\frac{t}{n}\right)\leq\exp(-t),$$

Similarly, for each of the other slices, we obtain that

$$\mathbf{P}\left(\exists\beta\in S_j\colon\ \mathbf{Z}(\beta)\geq\frac{\tilde{\lambda}_2(t)2^j}{n}+\frac{8h\phi^*(C'_{h,\phi^*}+\kappa\log(2n))}{3}\frac{t}{n}\right)\leq\exp(-t),$$

Using the union bound and the choice $t=\log p$, we obtain that

$$\begin{aligned}&\mathbf{P}\Big(\exists\beta\in\mathcal{B}\colon\ \mathbf{Z}(\beta)\geq\|\beta-\beta^*\|_1\tilde{\lambda}_2(t)\big|_{t=\log p}\\ &+\frac{8h\phi^*(C'_{h,\phi^*}+\kappa\log(2n))}{3}\frac{\log p}{n}\Big)\leq\frac{\log_2(\eta n\sqrt{p})}{p},\end{aligned}$$

which can be spelled out as

$$\begin{aligned}&\mathbf{P}\big(\exists\beta\in\mathcal{B}\colon\ \mathbf{Z}(\beta)\geq\|\beta-\beta^*\|_1\\ &\left[2\tilde{\lambda}_1+n^{-\kappa/2}\overline{C}_{\phi^*,h}C_X+\phi^*[\kappa\log(2n)+C'_{h,\phi^*}C_X]\sqrt{\frac{\log p}{n}}\right]\\ &+\frac{8h\phi^*(C'_{h,\phi^*}+\kappa\log(2n))}{3}\frac{\log p}{n}\Big)\leq\frac{\log_2(\eta n\sqrt{p})}{p}.\end{aligned} \tag{A20}$$

Putting together the pieces.

Combining (A3), (A4), (A6), (A19) and the argument of the previous section leading to (A20), we obtain that with the choice $\kappa=2$, the event

$$\begin{aligned}&\left\{\left|\{\nabla R_n(\beta)-\nabla R(\beta)\}^\top(\beta-\beta^*)\right|\right.\\ &\leq\|\beta-\beta^*\|_1\left(2C_0C_X\frac{\log(p+1)}{n}+4C_X\sqrt{\frac{\log(p+1)}{n}}+\frac{\overline{C}_{\phi^*,h}C_X}{n}\right.\\ &\quad+\phi^*[2\log(2n)+C'_{h,\phi^*}C_X]\sqrt{\frac{\log p}{n}}\\ &\quad\left.\left.+\frac{8h\phi^*(C'_{h,\phi^*}+2\log(2n))}{3}\frac{\log p}{n}\right)\ \forall\beta\in\mathcal{B}\right\}\end{aligned} \tag{A21}$$

occurs with probability at least $1-C_{\phi^*,h}/n-\log_2(\eta n\sqrt{p})/p$. Using Lemma 1 as indicated at the beginning of this proof, we can thus invoke Theorem A.1 with $\mu=\underline{\mu}/2$ and λ_ϵ equal to the term following $\|\beta-\beta^*\|_1$ in the right hand side of the inequality in (A21) to conclude the proof.

A.1 | Auxiliary Results

Theorem A.1. *A simplified version of Theorem 2.1 in (Elsener and van de Geer 2018). Let $\widehat{\beta}$ be a stationary point of (8) in an η-neighborhood C of β^*, and suppose that the risk R is $\underline{\mu}$-strongly convex over C. Let $\lambda_\epsilon>0$ and $\lambda_*\geq0$ such that for all $\beta\in C$ it holds that*

$$\left|\{\nabla R_n(\beta)-\nabla R(\beta)\}^\top(\beta-\beta^*)\right|\leq\lambda_\epsilon\|\beta-\beta^*\|_1+\frac{\underline{\mu}}{2}\|\beta-\beta^*\|_2^2+\lambda_*.$$

Let $\lambda>\lambda_\epsilon$, and define $\lambda_-=\lambda-\lambda_\epsilon$, $\lambda_+=\lambda+\lambda_\epsilon+\frac{1}{2}\lambda_-$. Then we have

$$\|\widehat{\beta}-\beta^*\|_1\leq2\left(\frac{1}{2\underline{\mu}}\frac{\lambda_+^2}{\lambda_-}s+\frac{\lambda_*}{\lambda_-}\right)$$

Lemma A.1. *Gradient and Hessian of R_n and R. We have*

$$\begin{aligned}\nabla R_n(\beta)&=\phi^*\frac{1}{n}\sum_{i=1}^n X_i\mu_i'(\beta)\{[\Psi(\mu_i(\beta)\phi^*)-\log(Y_i)]\\ &\quad-[\Psi((1-\mu_i(\beta))\phi^*)-\log(1-Y_i)]\}\\ \nabla R(\beta)&=\phi^*\mathbf{E}\big[X\mu'(X^\top\beta)\{[\Psi(\mu(X^\top\beta)\phi^*)-\Psi(\mu(X^\top\beta^*)\phi^*)]\\ &\quad-[\Psi((1-\mu(X^\top\beta))\phi^*)-\Psi((1-\mu(X^\top\beta^*))\phi^*)]\}\big]\\ \nabla^2R(\beta)&=\mathbf{E}\big[XX^\top\{\{\mu'(X^\top\beta)\}^2\phi^{*2}\{\Psi'(\mu(X^\top\beta)\phi^*)+\Psi'((1-\mu(X^\top\beta))\phi^*)\}\\ &\quad+\mu''(X^\top\beta)\phi^*\{[\Psi(\mu(X^\top\beta)\phi^*)-\Psi(\mu(X^\top\beta^*)\phi^*)]\\ &\quad-[\Psi((1-\mu(X^\top\beta))\phi^*)-\Psi((1-\mu(X^\top\beta^*))\phi^*)]\}\}\big]\end{aligned}$$

Proof. *The gradient $\nabla R_n(\beta)$ is obtained by direct calculation, noting that the digamma function Ψ equals the derivative of $\log\Gamma$. To obtain $\nabla R(\beta)=\mathbf{E}[\nabla R_n(\beta)]$, we use that $\mathbf{E}[\log(Y_i)|X_i]=\Psi(\mu_i(\beta^*)\phi^*)$ and $\mathbf{E}[\log(1-Y_i)|X_i]=\Psi((1-\mu_i(\beta^*))\phi^*)$, $1\leq i\leq n$, in light of properties of the log-Beta distribution (Gupta and Nadarajah 2004). The expression for $\nabla^2R(\beta)$ can be obtained from $\nabla R(\beta)$ by differentiating under the integral sign.* □

Lemma A.2. *Tail bound and second moments for log-Beta random variables. Let $Y\sim\mathsf{Beta}(\mu\phi,(1-\mu)\phi)$ for $\mu\in(c_h,1-c_h)$ with $c_h=1/(1+\exp(h))$ for $h>0$ and $\phi>0$. Consider $U=-\log(Y)$ and $W=-\log(1-Y)$. Then, for any $t\geq1$*

$$\begin{aligned}&\mathbf{P}(U>t)\leq C_{\phi,h}\exp(-t\mu\phi),\qquad&&\mathbf{P}(W>t)\leq C'_{\phi,h}\exp(-t(1-\mu)\phi),\\ &\mathbf{E}[U^2]\leq C''_{\phi,h},&&\mathbf{E}[W^2]\leq C'''_{\phi,h}\end{aligned}$$

for constants $C_{\phi,h},C'_{\phi,h},C''_{\phi,h},C'''_{\phi,h}>0$ depending only on ϕ and h.

Proof. *The probability density function of U is given by*

$$f_U(u)=\frac{\Gamma(\phi)}{\Gamma(\mu\phi)\Gamma((1-\mu)\phi)}[\exp(-u)^{\mu\phi-1}(1-\exp(-u))^{\phi(1-\mu)-1}]\exp(-u),\ \ u>0.$$

Consequently, $f_U(u)\leq\tilde{C}_{\phi,\mu}\exp(-u\mu\phi)$ *with* $\tilde{C}_{\phi,\mu}=\frac{\Gamma(\phi)}{\Gamma(\mu\phi)\Gamma((1-\mu)\phi)}$ *for any $u\geq1$. Therefore, for $t\geq1$* $\max\{(1-e^{-1})^{\phi(1-\mu)-1},1\}\leq\tilde{C}_{\phi,h}$

$$\mathbf{P}(U\geq t)=\int_t^\infty f_U(u)du\leq\tilde{C}_{\phi,h}\frac{1}{\mu\phi}\exp(-\mu\phi t)\leq C_{\phi,h}\exp(-\mu\phi t).$$

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It follows that

$$\begin{aligned}\mathbf{E}[U^2] = \int_0^\infty \mathbf{P}(U^2 \geq u)du &\leq 1 + \int_1^\infty \mathbf{P}(U \geq \sqrt{u})du \\ &\leq 1 + \int_1^\infty C_{\phi,h} \exp(-\mu\phi\sqrt{u})du \\ &\leq 1 + \int_1^\infty C_{\phi,h} 2z \exp(-\mu\phi z)dz \\ &\leq 1 + C'_{\phi,h} \frac{1}{\mu\phi} \int_0^\infty (\mu\phi) z \exp(-\mu\phi z)dz \\ &\leq 1 + C'_{\phi,h} \frac{1}{(\mu\phi)^2} \leq C''_{\phi,h},\end{aligned}$$

where we have applied the change of variables $z = \sqrt{u}$ *and used the fact that the expectation of an (Exponential) random variable with PDF* $z \mapsto \exp(-\lambda z), z, \lambda > 0$, *is given by* $1/\lambda$. *Moreover, we have set* $C'_{\phi,h} = 2C_{\phi,h}$.

The corresponding inequalities for W *can be established similarly, noting that* $1 - Y \sim \text{Beta}((1-\mu)\phi, \mu\phi)$. □

Lemma A.3. *General Lipschitz properties. Let* $f: \mathbb{R} \to \mathbb{R}$ *and* $g: \mathbb{R} \to \mathbb{R}$ *be bounded Lipschitz functions with Lipschitz constants* L_f *and* L_g, *respectively. Then* $f \circ g$ *and* $f \cdot g$ *are also Lipschitz functions with Lipschitz constants* $L_f L_g$ and $\|f\|_\infty L_g + L_f \|g\|_\infty$, *respectively.*

Proof. *Let* $x, x' \in \mathbb{R}$. *We have* $\begin{aligned}|f(g(x)) - f(g(x'))| &\leq L_f |g(x) - g(x')| \\ &\leq L_f L_g |x - x'|\end{aligned}$. *Regarding the second property, observe that*

□

$$\begin{aligned}|f(x)g(x) - f(x')g(x')| &= |(f(x) - f(x'))g(x) + f(x')(g(x) - g(x'))| \\ &\leq L_f \|g\|_\infty |x - x'| + L_g \|f\|_\infty |x - x'|.\end{aligned}$$

Lemma A.4. *Lipschitz and boundedness properties. With some abuse of notation, let* $\mu(\beta) = \mu(x^\top\beta) = \exp(x^\top\beta)/\{1 + \exp(x^\top\beta)\}$ and $\mu'(\beta) = \frac{d\mu(z)}{dz}\Big|_{z = x^\top\beta}$. *Let further* $\Psi(z) = \frac{d}{dz}\log\Gamma(z), z > 0$, *be the Digamma function and let* Ψ' and Ψ'' *denote its first and second derivative, respectively, restricted to the interval* $[c_h\phi, \phi)$ *with* $c_h = (1 + \exp(h))^{-1}$ *for* $h > 0$. *We have*

$$\begin{aligned}&(i)\ 0 \leq \mu'(\beta) \leq \frac{1}{4} \text{ and } |\mu(\beta) - \mu(\beta')| \leq \frac{1}{4}|x^\top(\beta - \beta')|, \\ &\quad |\mu''(\beta)| \leq \frac{1}{4},\ |\mu'''(\beta)| \leq \frac{3}{8}. \\ &(ii)\ \|\Psi\|_\infty \leq C_{h,\phi},\ L_\Psi = \Psi'(c_h\phi) = C'_{h,\phi} < \infty, \\ &\quad L_{\Psi'} = |\Psi''(c_h\phi)| = C''_{h,\phi} < \infty,\end{aligned}$$

where L_Ψ *and* $L_{\Psi'}$ *denotes the Lipschitz constants of* Ψ and Ψ', *respectively, when restricted to* $[c_h\phi, \phi)$.

Proof. *One computes that* $\mu'(\beta) = \mu(\beta) \cdot (1 - \mu(\beta))$, *which yields the first part in (i). Furthermore, we have* $\mu''(\beta) = \mu'(\beta)(1 - 2\mu(\beta))$ *whose range is* $[-1/4, 1/4]$. *Similarly,* $\mu'''(\beta) = \mu'(\beta)(1 - 2\mu(\beta))^2 - 2\mu'(\beta)^2$ *so that* $\mu'''(\beta) \leq \frac{1}{4} + \frac{2}{16} = \frac{3}{8}$. *Regarding part (ii), observe that* Ψ *is monotonically increasing with* $\lim_{z\to\infty}\Psi(z) = \infty$, *and an asymptote at zero such that* $\lim_{z\to 0}\Psi(z) = -\infty$. *Therefore,* $\|\Psi\|_\infty = \max\{|\Psi(c_h\phi)|, \Psi(\phi)\} = C_{h,\phi}$. *We also note that* Ψ' *is non-negative and monotonically decreasing with an asymptote at zero such that* $\lim_{z\to 0}\Psi'(z) = \infty$. Since $c_h < 1$, *we have* $L_\Psi = \max_{z\in[c_h\phi,\phi)}\Psi'(z) = \Psi'(c_h\phi)$. *Moreover,* Ψ'' *is non-positive and monotonically increasing with* $\lim_{z\to 0}\Psi''(z) = -\infty$ *and* $\lim_{z\to\infty}\Psi''(z) = 0$, *so that* $L_{\Psi'} = \max_{z\in[c_h\phi,\phi)}|\Psi''(z)| = |\Psi''(c_h\phi)|$. □

Lemma A.5. *Contraction principle (van de Geer 2008, Theorem A.3). Let* $z_1, \ldots, z_n$ *be non-random elements of some space* $\mathcal{Z}$, *and let* $\mathcal{F}$ *be a class of real-valued functions on* $\mathcal{Z}$. *Consider Lipschitz functions* $\gamma_i: \mathbb{R} \to \mathbb{R}$, i.e., $|\gamma_i(s) - \gamma_i(\tilde{s})| \leq |s - \tilde{s}|\ s, \tilde{s} \in \mathbb{R}, 1 \leq i \leq n$. *Let* $\epsilon_1, \ldots, \epsilon_n$ *be i.i.d. Rademacher variables. Then for any function* $f^*: \mathcal{Z} \to \mathbb{R}$, *we have*

$$\mathbf{E}\left[\sup_{f\in\mathcal{F}}\left|\sum_{i=1}^n \epsilon_i\{\gamma_i(f(z_i)) - \gamma_i(f^*(z_i))\}\right|\right] \leq 2\mathbf{E}\left[\sup_{f\in\mathcal{F}}\left|\sum_{i=1}^n \epsilon_i\{f(z_i) - f^*(z_i)\}\right|.\right]$$

Lemma A.6. *Concentration inequality for empirical processes (van de Geer 2016 Corollary 16.1). Let* $\{Z_i\}_{i=1}^n$ *be independent random variables with range* $\mathcal{Z}$, *and let* $\mathcal{F}$ *be a class of real-valued functions on* $\mathcal{Z}$ *such that* $\|f\|_\infty < K < \infty$ *and* $\mathbf{E}[f(Z_i)] = 0, 1 \leq i \leq n$, *for all* $f \in \mathcal{F}$. *Suppose further that* $\frac{1}{n}\sum_{i=1}^n \sup_{f\in\mathcal{F}}\mathbf{E}[f^2(Z_i)] \leq R^2 < \infty$, *and let* $\mathbf{Z} = \left|\sup_{f\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n f(Z_i)\right|$. *Then for all* $t > 0$

$$\mathbf{P}\left(\mathbf{Z} \geq 2\mathbf{E}[\mathbf{Z}] + R\sqrt{\frac{t}{n}} + \frac{4tK}{3n}\right) \leq \exp(-t).$$

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